Letters to the Editor

Diamond Burr Polishing

Dear Editor:
The authors report that outcomes after diamond burr polishing are better than those after phototherapeutic keratectomy (PTK) when treating recurrent corneal erosions (RCEs).[1] This is good news to all of the ophthalmologists and patients who do not have access to an excimer laser.

The unfavorable results reported after excimer laser treatments, though disappointing, are nonetheless intriguing. As the authors mention in their discussion, PTK has been generally effective in the treatment of RCEs. Our own experience with PTK has also been positive,[2] even when treating recurrent erosions in bullous keratopathy.[3]

I have comments on two problems mentioned in this article:

1. We have encountered mild corneal haze in only one patient after treatment. In that case no subjective reduction of visual function was observed by the patient. It should be mentioned that some patients with RCE do have some haze before treatment. Slit-lamp examination of the cornea can reveal where corneal erosions will start. The authors do not mention if they looked for corneal haze preoperatively. The fact that this study is retrospective and not masked weakens the authors' conclusions.
2. It is common and effective practice in the treatment of RCEs to debride most of the epithelium using a sponge alone. This is effective because the adhesion between the epithelium and the underlying Bowman's layer is very loose, not only where a loose epithelium is being observed, but also many times on the whole corneal surface. Because most commercial excimer lasers have a maximal ablation diameter of 10 mm, it is extremely important to treat the periphery of the erosion. (The VISX, which had been used by these authors, has a maximal ablation diameter of 9 mm.) The authors should mention the diameter they used, and whether they extended the area treated to include the periphery.

We use computer-controlled ablation in the central 10 mm and treat the periphery manually. Our technique is not only in a nonoverlapping manner, as described by the authors, but in an overlapping manner as well, including an overlapping of the epithelium. This assures equivalent good adhesion in the periphery. We have had no recurrences since using this technique.

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Author reply

Dear Editor:
Dr. Schipper writes that we concluded "that outcomes after diamond burr polishing are better than those after phototherapeutic keratectomy (PTK)." We did not state this. To paraphrase our conclusions, we stated that both treatments are effective; however, diamond burr polishing, being simpler and less expensive, may have advantages over PTK.

Dr. Schipper notes correctly that patients may have preexisting corneal haze before a diamond burr or PTK procedure. We did not systematically look for preoperative corneal haze, as many patients had their procedure during an episode of recurrent erosion where haze would be difficult to quantify. As noted in the article, we did not find a significant difference in postoperative haze between the two groups.

Dr. Schipper discussed his technique of PTK, where he uses a 10-mm ablation. He states that the VISX laser (VISX Inc, Santa Clara, CA) we used has a 9-mm maximal ablation capability. We clearly discuss the exact PTK technique in the second paragraph of Materials and Methods. The VISX laser models we used (20/20 B and Star S2, as stated in that section) have maximal PTK ablation diameters of 6.0 mm. We described our technique of debriding the loose epithelium and treating with the laser. If the epithelial defect was 6 mm in diameter or smaller, we used a single large-zone ablation of 6 mm. If the epithelial defect was larger than 6 mm in diameter, then the central 6-mm area was treated and then masked with a 6-mm-diameter filter paper and the periphery was treated with nonoverlapping 3- to 4-mm-diameter ablations. Consequently, the entire epithelial defect would be treated just once for 5 μm. The outer treatments did overlap the peripheral intact epithelium.

We continue to use both diamond burr polishing and PTK as part of our armamentarium in the management of recurrent erosion syndrome.

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Corneal Refractive Power after Myopic LASIK

Dear Editor:
The recent article by Hamed et al[1] addresses an emerging problem that many ophthalmologists will be facing in the relatively near future. It has two main conclusions: (1) when manual keratometry and topography are used for intraocular lens (IOL) calculations after myopic LASIK, the power is underestimated, and (2) there is a direct, linear relationship between the degree of corneal refractive surgery and IOL power error as evidenced by regression analysis. This study also makes several important recommendations, the most notable of which is reducing the keratometry readings by 0.24 diopters (D) per diopter of refractive correction and

using this calculated corneal power to determine IOL power. The idea of a "fudge factor" is certainly both desirable and practical. There are, however, several points about this study that need further scrutiny. The authors make the assumption that using the "historical method" leads to the most accurate corneal power. This is the gold standard by which all the other four methods are compared. In the historical method, as described by the authors, the change in spherical equivalent induced by LASIK at the corneal plane is subtracted from post-LASIK keratometry. In doing so, the authors do not take into consideration that there is not a one-to-one relationship between corneal power and refractive power. In other words, when the refraction at the corneal plane is changed by 1 D, the readings do not change by the same amount.

We draw attention to a similar published study that addresses this issue.[2] We performed a study of IOL calculation after both myopic and hyperopic LASIKs and arrived at similar conclusions. Our data verified a linear relationship between the amount of LASIK correction and the IOL power error. In addition, we also found that to compensate for this error after myopic LASIK, the pre-LASIK IOL power had to be increased by a factor of 0.7 for every diopter of myopic correction induced by LASIK. In the same study, the use of the historical method as described by Hamed et al[1] was also analyzed. Our results indicated that although the use of the method increased accuracy, after LASIK for myopia of more than 2 D, this method still underestimated the IOL power.

What exactly is the source of error in determining true corneal power after keratorefractive surgery? The answer most commonly cited is the alteration in the relationship between the anterior and posterior corneal surface that results in a change in refractive index. One may ask whether excimer ablation of the cornea inherently changes the corneal stroma, thereby partially altering the refractive index. There may be some evidence supporting this. A study by Patel et al[3] demonstrated that different layers of the cornea have slightly different refractive indices. Perhaps selective removal of layers of the cornea by the excimer laser results in a change in refractive index. In Hamed and colleagues' study,[1] when using the Gaussian formula, the authors employed a refractive index of 1.376 for the anterior surface of the cornea. However, this value is true only of a native cornea and not for an ablated cornea after selective removal of particular layers. This may partially explain why this approach, though taking corneal thickness into account, may still result in errors.

In summary, the authors' conclusions and recommendations appear to verify the results of previously published studies.[2,4] The most accurate method of calculating IOL power at this point remains use of refractive change induced by LASIK and a fudge factor to compensate for the change in refractive index of the cornea. In doing so, the relationship between corneal power and refraction needs to be considered.

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Author reply

Dear Editor:

We enjoyed reading Drs. Feiz and Mannis' letter and welcome the study of new approaches for calculating intraocular lens (IOL) power in patients who have undergone LASIK and photorefractive keratectomy. However, we do not feel that their letter accurately characterizes their method or the current state of knowledge regarding this challenging problem.

In their critique of the historical (more commonly referred to as "clinical history") method, the authors state that "there is not a one-to-one relationship between corneal power and refractive power." This misses the point. The clinical history method is based on the obvious fact that the refractive change induced by the excimer laser procedure occurs in the cornea. Therefore, the change in refractive error is simply vertexed back to the corneal plane, which is where this change took place.

We are surprised by their comment that the clinical history method "still underestimated the IOL power." As they state in their article, none of these methods has been tested in a clinical series. In their method, they divide the LASIK-induced refractive change by 0.7 and add this to the IOL power that was calculated for the pre-LASIK eye. A potential flaw in this approach is that it could result in overestimation of postoperative IOL power, resulting in unanticipated postoperative myopia. Therefore, any comparative evaluation of these methods is purely speculation, pending further clinical study.

In their critique of our use of the Gaussian formula, they comment that the refractive index of 1.376 is not likely to be accurate. We disagree, because 1.376 represents the refractive index of the air–tear interface and should not be affected by changes in refractive indices of various layers of the cornea.

Finally, we point out that a unique advantage of the method we presented in our article (adjusting postoperative corneal measurements according to the amount of LASIK-induced refractive change) is that it requires only one datum from before LASIK—the amount of myopia. This can often be provided by patients even though they cannot obtain copies of preoperative keratometry or topography measurements, which are required for both the clinical history and Drs. Feiz and Mannis' methods.